Exercise 6: Cursors

Aim

To use MySQL cursors to process records one by one for generating customer statements, applying annual maintenance fees, and updating loan interest rates.

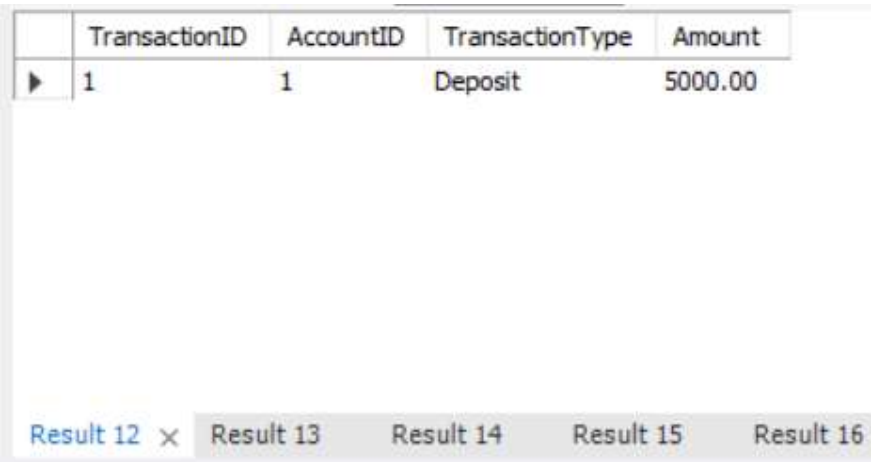
--Utilizing the **same database** from exercise 1, 3, 4 and 5

--**Question:** Write a PL/SQL block using an explicit cursor **GenerateMonthlyStatements** that retrieves all transactions for the current month and prints a statement for each customer.

```
INSERT INTO Transactions VALUES
(101,1,'Deposit',5000),
(102,2,'Withdrawal',3000),
(103,3,'Deposit',7000);
DELIMITER $$

CREATE PROCEDURE GenerateMonthlyStatements()

BEGIN

DECLARE Done BOOLEAN DEFAULT FALSE;

DECLARE T_ID INT;
DECLARE A_ID INT;
DECLARE T_Type VARCHAR(20);
DECLARE T_Amount DECIMAL(10,2);

DECLARE cur CURSOR FOR
```

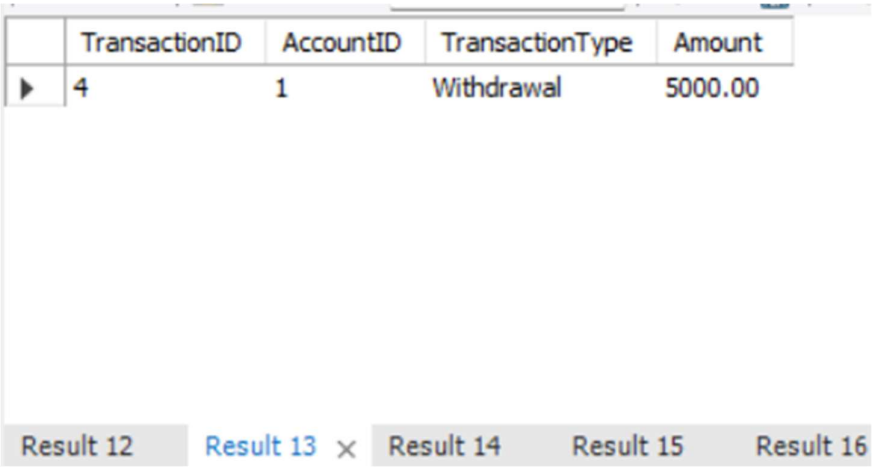
```
SELECT TransactionID,
      AccountID,
      TransactionType,
      Amount
FROM Transactions;

DECLARE CONTINUE HANDLER
FOR NOT FOUND
SET Done = TRUE;

OPEN cur;

read_loop: LOOP

FETCH cur
INTO T_ID,
    A_ID,
    T_Type,
    T_Amount;

IF Done THEN
LEAVE read_loop;
END IF;

SELECT
T_ID AS TransactionID,
A_ID AS AccountID,
T_Type AS TransactionType,
T_Amount AS Amount;
```

```
END LOOP;

CLOSE cur;

END $$

DELIMITER ;

CALL GenerateMonthlyStatements();
```



	TransactionID	AccountID	TransactionType	Amount
▶	1	1	Deposit	5000.00

Result 12 × Result 13 Result 14 Result 15 Result 16



	TransactionID	AccountID	TransactionType	Amount
▶	4	1	Withdrawal	5000.00

Result 12 Result 13 × Result 14 Result 15 Result 16

	TransactionID	AccountID	TransactionType	Amount
▶	101	1	Deposit	5000.00

Result 12 Result 13 Result 14 × Result 15 Result 16

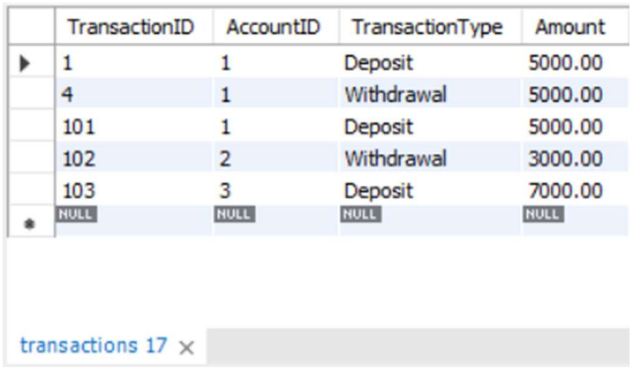
	TransactionID	AccountID	TransactionType	Amount
▶	102	2	Withdrawal	3000.00

Result 12 Result 13 Result 14 Result 15 × Result 16

	TransactionID	AccountID	TransactionType	Amount
▶	103	3	Deposit	7000.00

Result 12 Result 13 Result 14 Result 15 Result 16 ×

SELECT * FROM transaction;



TransactionID	AccountID	TransactionType	Amount
1	1	Deposit	5000.00
4	1	Withdrawal	5000.00
101	1	Deposit	5000.00
102	2	Withdrawal	3000.00
103	3	Deposit	7000.00
NULL	NULL	NULL	NULL

transactions 17 ×

--**Question:** Write a PL/SQL block using an explicit cursor **ApplyAnnualFee** that deducts an annual maintenance fee from the balance of all accounts.

```
DELIMITER $$

CREATE PROCEDURE ApplyAnnualFee()

BEGIN

DECLARE Done BOOLEAN DEFAULT FALSE;

DECLARE A_ID INT;

DECLARE cur CURSOR FOR

SELECT AccountID
FROM Account;

DECLARE CONTINUE HANDLER
FOR NOT FOUND
SET Done = TRUE;
```

```
OPEN cur;

read_loop: LOOP

FETCH cur
INTO A_ID;

IF Done THEN
LEAVE read_loop;
END IF;

UPDATE Account

SET Balance = Balance - 100

WHERE AccountID = A_ID;

END LOOP;

CLOSE cur;

END $$

DELIMITER ;
```

SELECT * FROM Account;

AccountID	CustomerID	AccountType	Balance
1	101	Savings	45400.00
2	102	Savings	35200.00
3	103	Savings	15050.00
4	104	Current	40300.00
5	105	Savings	25150.00
NULL	NULL	NULL	NULL

CALL ApplyAnnualFee();

SELECT * FROM Account;

AccountID	CustomerID	AccountType	Balance
1	101	Savings	45300.00
2	102	Savings	35100.00
3	103	Savings	14950.00
4	104	Current	40200.00
5	105	Savings	25050.00
NULL	NULL	NULL	NULL

--**Question:** Write a PL/SQL block using an explicit cursor **UpdateLoanInterestRates** that fetches all loans and updates their interest rates based on the new policy.

```
DELIMITER $$

CREATE PROCEDURE UpdateLoanInterestRates()

BEGIN

DECLARE Done BOOLEAN DEFAULT FALSE;

DECLARE C_ID INT;

DECLARE cur CURSOR FOR
```

```
SELECT CustomerID
FROM Customer;

DECLARE CONTINUE HANDLER
FOR NOT FOUND
SET Done = TRUE;

OPEN cur;

read_loop: LOOP

FETCH cur
INTO C_ID;

IF Done THEN
LEAVE read_loop;
END IF;

UPDATE Customer

SET LoanInterestRate = LoanInterestRate - 0.5

WHERE CustomerID = C_ID;

END LOOP;

CLOSE cur;

END $$

DELIMITER ;

SELECT CustomerID,  CustomerName,   LoanInterestRate  FROM Customer;
```

CustomerID	CustomerName	LoanInterestRate
101	Krishna	2.50
102	Rahul	10.00
103	Priya	1.50
104	Arun	11.00
105	Sneha	2.00
106	Karan	9.50
NULL	NULL	NULL

```
CALL UpdateLoanInterestRates();
```

```
SELECT CustomerID,  CustomerName,   LoanInterestRate  FROM Customer;
```

CustomerID	CustomerName	LoanInterestRate
101	Krishna	2.00
102	Rahul	9.50
103	Priya	1.00
104	Arun	10.50
105	Sneha	1.50
106	Karan	9.00
NULL	NULL	NULL